Success rate across the DD transition, by depth, data and width

N=48, d=5, q=4, FALL ensemble -- 13 cells, 400 instances per density, commit cd97030, jobs 50760297 / 50760252

H1 The transition is a transition -- CONFIRMED as shape, over-run at the deep end

Every cell that learns the task at all produces a monotone breakdown curve, not a flat line: solve falls from ~1.0 at rho=0.94 to <0.1 at the ensemble floor rho=0.333 for 9 of the 11 learning cells. The pre-registered falsifier (flat to within 0.1, or rho_50<0.40 everywhere) did not fire. But the sub-prediction rho_50>0.50 FAILS for the two strongest cells: L=12 at 120k gives 0.460, and L=12 at 240k never drops to 0.5 at all -- it is still at 0.755 where the fall ensemble runs out of removable hints. At N=48 the best model now outruns the generator, which is a statement about N=48, not about the model.

H2 Depth buys the tail -- CONFIRMED at matched data, not separable at 30k

At 120k examples rho_50 = 0.744 / 0.570 / 0.460 for L = 4 / 8 / 12, monotone, a gap of 0.284 against a pre-registered bar of 0.10. The curves agree to 0.030 at rho=0.94 (bar 0.03) and separate only below rho~0.8, exactly as predicted. At 30k the L ordering inverts by 0.157 in rho_50 -- smaller than the 0.154 seed spread measured at L=8, so it is not separable at this budget.

H3 Data helps most in the tail -- direction confirmed, size separable only at L=12

Going 30k -> 240k distinct examples at equal 375k gradient steps moves tail solve (rho<=0.5) by +0.86 at L=12, +0.06 at L=8 and +0.00 at L=4; the easy end (rho>=0.8) moves +0.14 / +0.00 / -0.00. Only the L=12 effect clears s_L8. The unregistered finding is the INTERACTION: depth pays only if it is fed. The 30k cells overfit as the pre-registration warned they might (val-train gap +0.14 at L=8, +0.24 at L=12, against -0.01 at 120k/240k).

H4 It is depth, not parameters -- CONFIRMED, and this is the load-bearing cell

L=4 with ff=4096 has 9.60M parameters against L=12/ff=1024's 9.61M, a 0.15% match. Its rho_50 is 0.748, which is 0.004 from the 3.30M L=4 model (bar <=0.03) and +0.288 above L=12 (bar >=+0.08). Its whole curve lies on top of the narrow L=4 curve to within 0.042 everywhere. 2.91x the parameters at fixed depth buys nothing. Every depth reading in findings 11/14/22/33/38 stands as a depth reading.

All four hypotheses and the scoring rule were written into the sbatch header before submission. Scoring rule, verbatim: any level comparison across cells is read against s_L8 at the same rho; where the effect is smaller the statement is 'not separable at this budget', not a result with a p-value. Curve shape within one checkpoint is paired and does not need this caveat.

What the model actually sees

Every token is one categorical id into a 58-symbol vocabulary (N=48, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING
	BEGIN_ROUND SOLVED UNRESOLVED
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	COLOR_0 COLOR_1 COLOR_2 COLOR_3
	the q colours (token name COLOR_n, prose 'colour')
10-57	N0 N1 ... N47
	one id per node, no structure in the id

GRAMMAR

BEGIN_HINTS	(NODE COLOR)*
BEGIN_GRAPH	(NODE (NODE){degree})*
BEGIN_REASONING	(BEGIN_ROUND (NODE COLOR)+)*

SOLVED

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image:
node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 401 TOKENS

HINTS	BEGIN_HINTS N0 COLOR_0 N1 COLOR_2 N2 COLOR_3 N10 COLOR_1 N15 COLOR_1 N17 COLOR_3 ... (37 tokens)
GRAPH	BEGIN_GRAPH N0 N16 N8 N38 N34 N17 N1 N13 N27 N34 N28 N10 ... (289 tokens)
REASONING	BEGIN_REASONING BEGIN_ROUND N9 COLOR_3 N14 COLOR_2 N16 COLOR_2 N32 COLOR_0 N34 COLOR_1 N44 ... (75 tokens)

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 13 rounds, 30 deductions,
60 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000018, FALL-colouring ensemble (every node sees all q-1 other colours, so no node is permanently undeducible).

One real reasoning trace

18 hints of 48 nodes, then 13 rounds of simultaneous deduction. Hints: N0=COLOR_0 N1=COLOR_2 N2=COLOR_3 N10=COLOR_1 N15=COLOR_1 N17=COLOR_3 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND      N9 COLOR_3 N14 COLOR_2 N16 COLOR_2 N32 COLOR_0 ...
  6 deduced
BEGIN_ROUND      N42 COLOR_3 N45 COLOR_1
  2 deduced
BEGIN_ROUND      N13 COLOR_0 N20 COLOR_0 N37 COLOR_2
  3 deduced
BEGIN_ROUND      N21 COLOR_2 N24 COLOR_3 N31 COLOR_2
  3 deduced
... 9 further rounds

SOLVED
```

WHY ONE DEDUCTION FIRES

node N9, neighbours N25 N10 N29 N31 N18

their colours known before this round:

```
[0, 1, 2]  (3 = q-1 of the 4)

(3 known neighbours, 3 DISTINCT colours)

still unknown: N25 N31
```

only colour left: 3

```
so the trace emits  N9 COLOR_3
```

The model is never shown this reasoning -- only the node/colour pair.

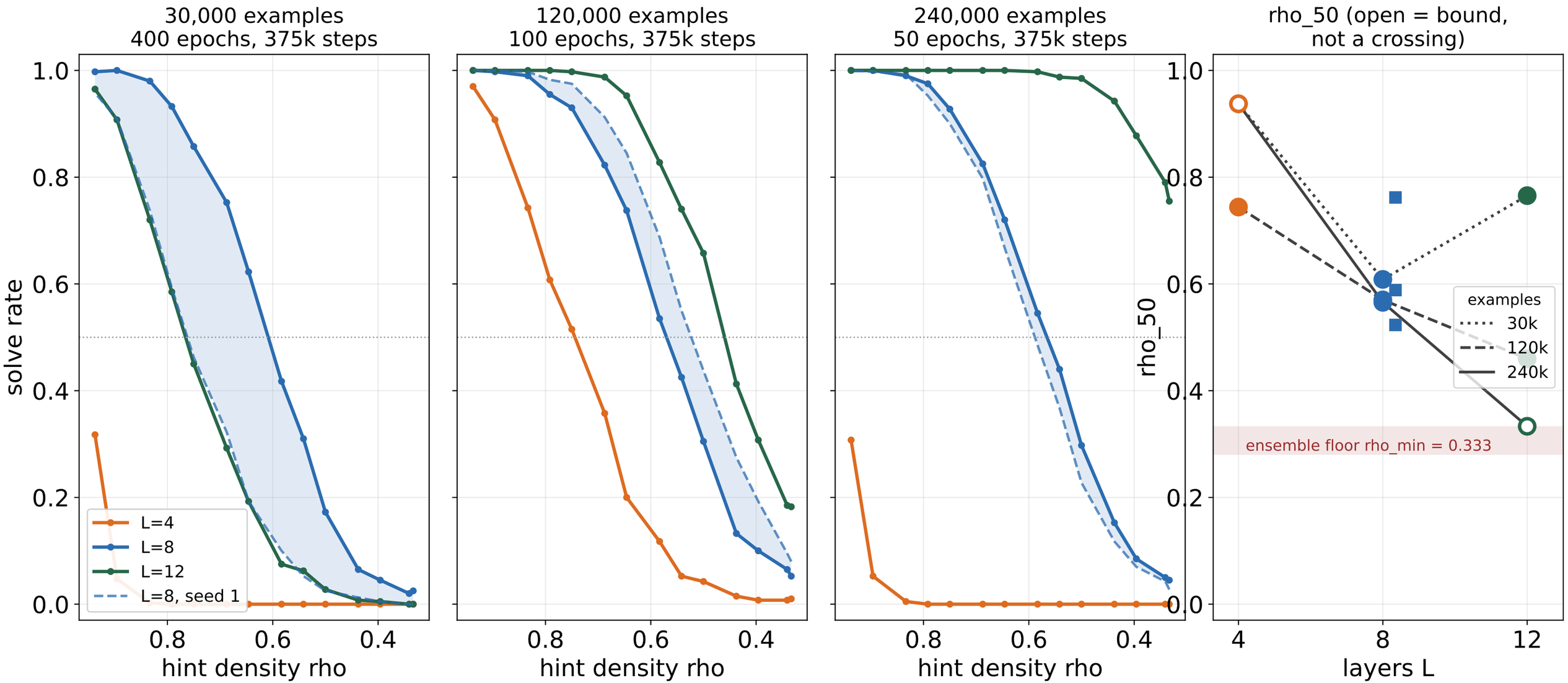
A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

N=48, degree 5, 4 colours, --shuffle-adj --fall: the task all 13 cells in this deck were trained on. rho=0.33 is the deep end of the eval grid, where only the L=12 c

T=13 rounds, the 90th percentile of 40 draws from this config (range 1-16) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

The transition, one panel per data budget

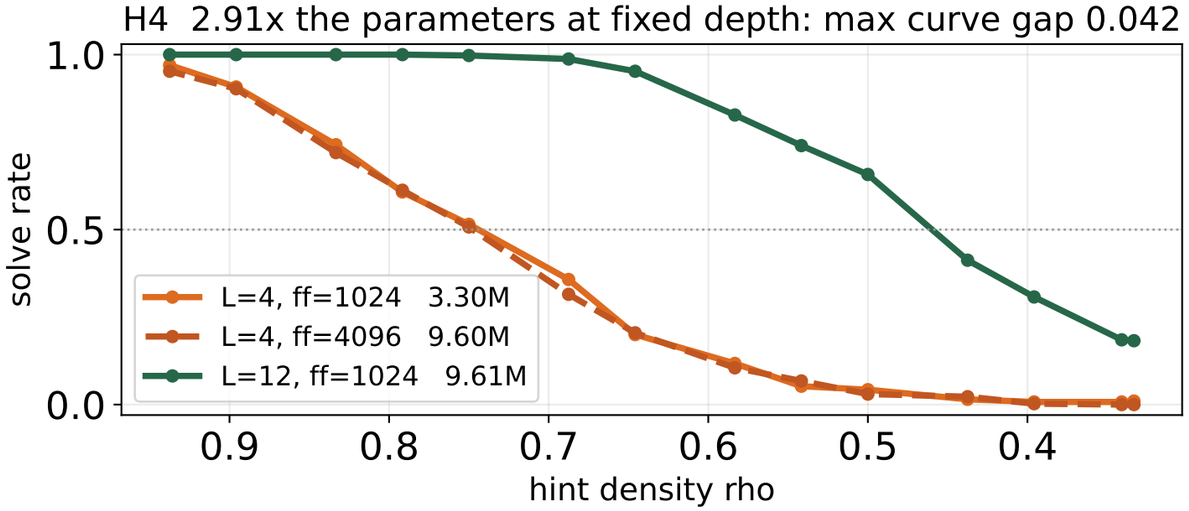
solve rate vs realized hint density; every curve is the SAME 400 instances per density

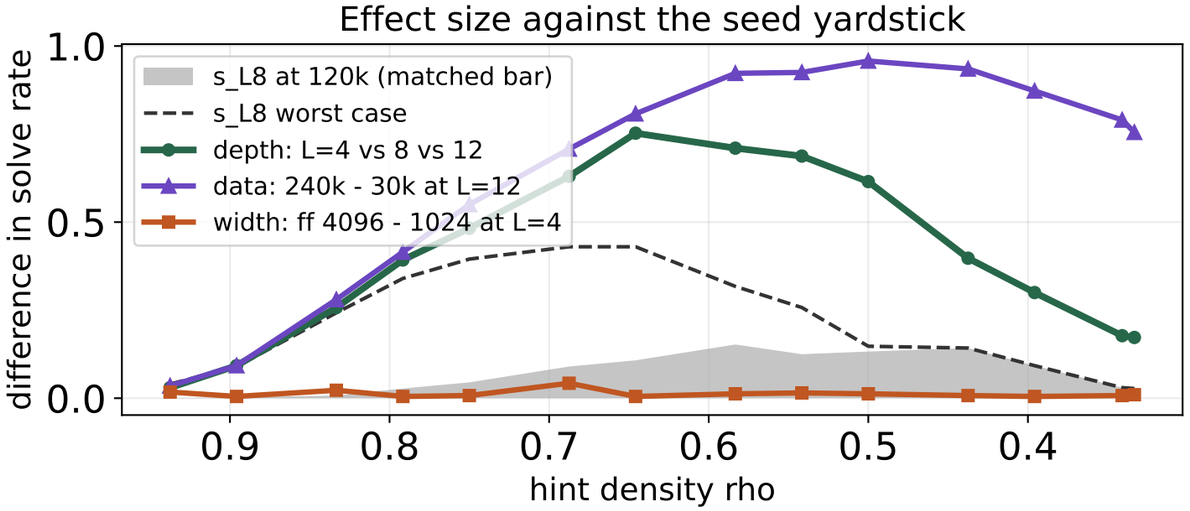


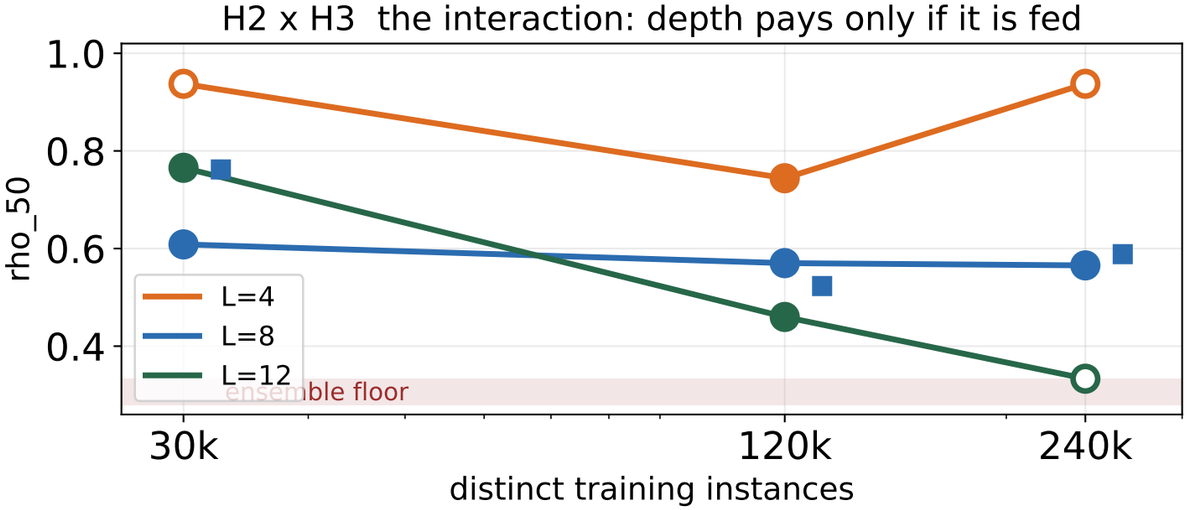
Open markers never cross 0.5 inside the reachable range: L=4 at 30k and 240k because the run never left its loss plateau (next page), L=12 at 240k because it still solves 75.5% where the ensemble runs out of removable hints. Squares are the L=8 seed-1 replicates, and the shaded band between the two L=8 seeds is the yardstick every cross-cell claim here is read against.

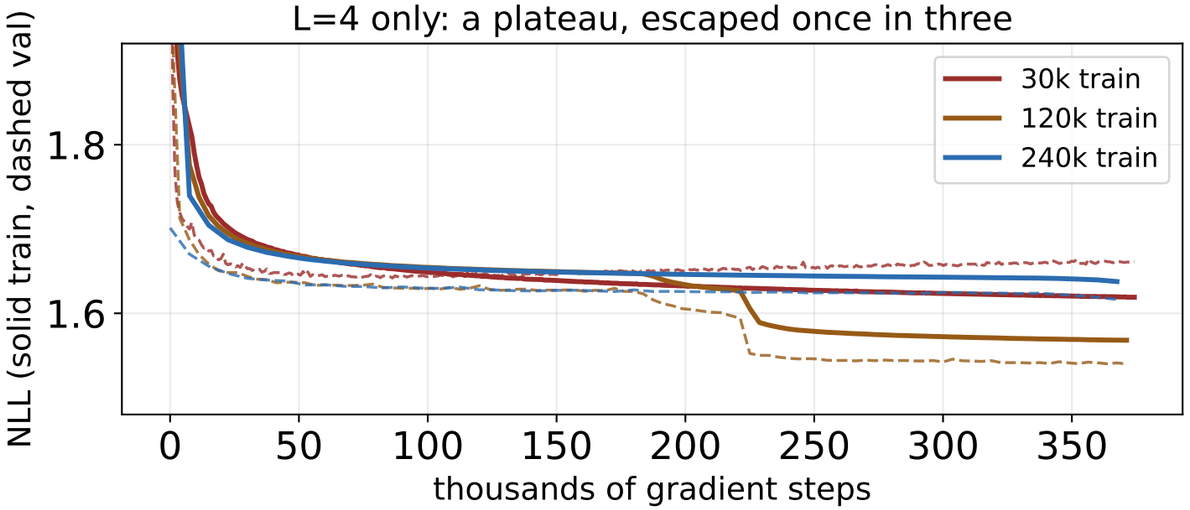
What actually moves the curve: depth yes, data yes, parameters no

equal 375k gradient steps everywhere, so the data axis varies DISTINCT examples with optimization held fixed









An effect is claimed only where it clears the grey band. Depth clears it from rho=0.79 down, peaking at 0.752 at rho=0.65; data at L=12 clears it everywhere below rho=0.83; width never clears it anywhere, its largest excursion being 0.042 between two models whose parameter counts differ by 0.15%. Bottom right: all three L=4 cells settle near NLL 1.64 and the 120k one fell off that plateau around step 230k, with the 240k cell cut off at 375k steps at exactly the NLL the 120k cell had when it escaped.

Every cell

N=48, d=5, q=4, fall ensemble, d_model=256, 4 heads, lr 1e-3 constant, batch 32, 375k steps, --shuffle-adj; commit cd97030

task	L	ff	params	n_train	ep	seed	train	val	gap	0.94	0.75	0.65	0.50	0.33	rho_50	
0	4	1024	3.30M	30,000	400	0	1.619	1.661	+0.042	0.318	0.000	0.000	0.000	0.000	0.938	>top
1	4	1024	3.30M	120,000	100	0	1.568	1.540	-0.028	0.970	0.515	0.200	0.043	0.010	0.744	
2	4	1024	3.30M	240,000	50	0	1.638	1.616	-0.021	0.307	0.000	0.000	0.000	0.000	0.938	>top
12	4	4096	9.60M	120,000	100	0	1.554	1.542	-0.012	0.953	0.507	0.205	0.030	0.000	0.748	
3	8	1024	6.45M	30,000	400	0	1.440	1.580	+0.139	0.998	0.858	0.623	0.172	0.025	0.608	
9	8	1024	6.45M	30,000	400	1	1.458	1.611	+0.154	0.958	0.463	0.193	0.025	0.000	0.762	
4	8	1024	6.45M	120,000	100	0	1.506	1.500	-0.006	1.000	0.930	0.738	0.305	0.052	0.570	
10	8	1024	6.45M	120,000	100	1	1.503	1.497	-0.006	1.000	0.975	0.845	0.438	0.080	0.523	
5	8	1024	6.45M	240,000	50	0	1.514	1.503	-0.011	1.000	0.927	0.720	0.297	0.045	0.565	
11	8	1024	6.45M	240,000	50	1	1.513	1.502	-0.012	1.000	0.900	0.667	0.228	0.028	0.588	
6	12	1024	9.61M	30,000	400	0	1.407	1.650	+0.243	0.965	0.450	0.193	0.028	0.000	0.765	
7	12	1024	9.61M	120,000	100	0	1.495	1.489	-0.006	1.000	0.998	0.953	0.657	0.182	0.460	
8	12	1024	9.61M	240,000	50	0	1.490	1.484	-0.006	1.000	1.000	1.000	0.985	0.755	0.333	<floor

	0.95	0.75	0.65	0.50	0.33
requested rho					
realized rho	0.938	0.750	0.646	0.500	0.333
mean DD rounds T	1.15	2.79	4.08	7.18	13.79

'gap' is final val NLL minus final train NLL. The three 30k cells are the only ones that overfit, and they overfit more the deeper they are (+0.042 at L=4, +0.140 at L=8, +0.243 at L=12) -- the pre-registration named this as a thing to watch and said it would be part of the answer rather than a reason to drop the cells. Note that val NLL does not rank the cells: L=12/30k has the WORST val NLL of all 13 (1.650) and still solves 96.5% at rho=0.94.

Controls, and what this measurement cannot carry

pre-registered checks, scored after the fact against the recorded CSVs

PASS pairing, and the ensemble floor is labelled not fitted

All 13 curves see an identical realized density at each of the 14 requested rho values (verified), so every within-grid comparison is paired point for point. Exactly one requested point (0.33) lands on the ensemble floor rho_min=0.3333, as designed -- the grid is 14 distinct realized densities, not one density measured several times. dd_mean_T runs 1.15 -> 13.79 rounds across it.

FAIL outcome accuracy and truncation, in the direction that UNDER-reports the deep cells

The bars were outcome_accuracy >= 0.99 and truncated_rate <= 0.01 at every density with solve > 0.1. 8 of 115 live bins miss them, worst case 0.9475 and 0.0275. All 8 belong to the two L=12 cells at the three deepest densities. The cause is the context window: max_len is 414 tokens and the longest true fall trace at N=48 is 408, so a model that emits even a few spare tokens at rho=0.33 is cut off mid-trace. solve_rate() separates this from reasoning failure by design, and the effect is bounded by 0.028 -- far below the 0.6-0.8 depth effects -- so no conclusion turns on it, but the L=12 tail numbers are a lower bound and the cap must be raised before N=96.

CAVEAT two of the three L=4/ff=1024 cells never left their loss plateau

L=4 is at the edge of trainability on this task. Its three cells all plateau near NLL 1.64 and exactly one escaped inside 375k steps. The L=4 row therefore measures trainability, not capacity, and the honest reading of H2 and H3 at L=4 is 'no measurement'. The wide L=4 cell trained normally, which is what makes H4 readable at all.

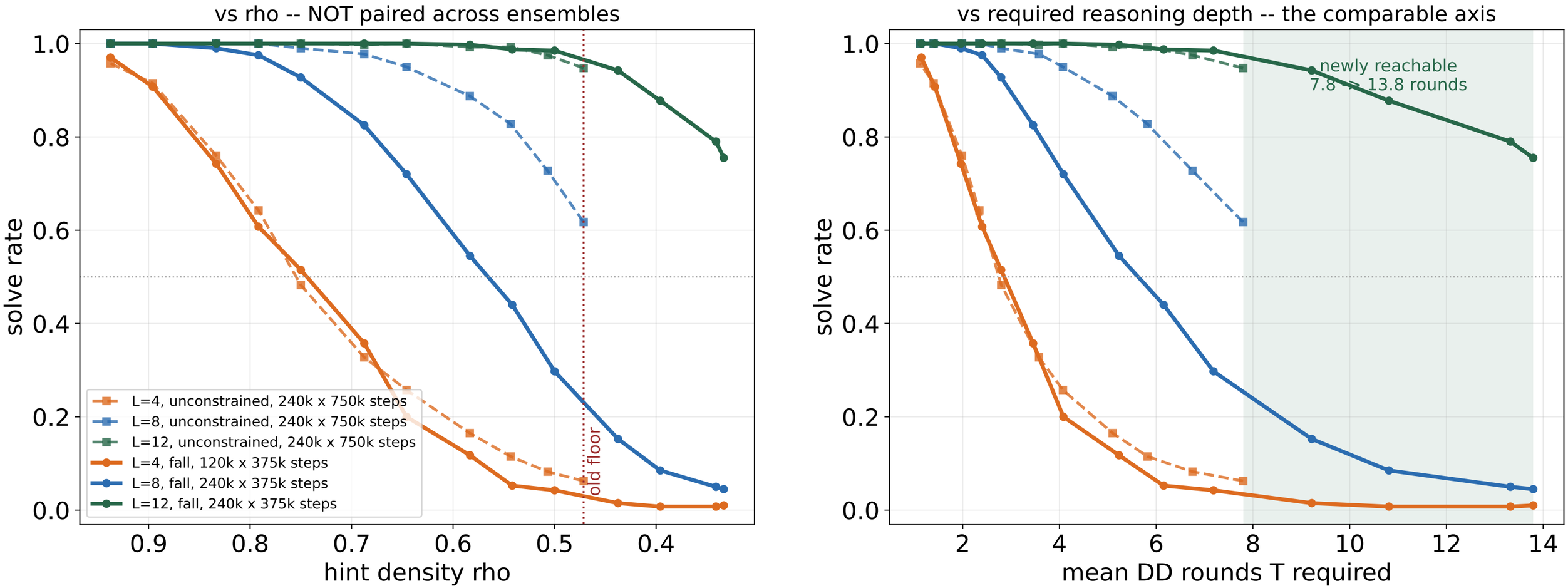
WHAT THIS DOES NOT SHOW

One architecture family, one N, one learning rate, one seed per cell off the L=8 column. s_L8 is itself estimated from three seed pairs, so the yardstick has its own error bar. Nothing here says the network is running DD -- that is the SAE/ablation question, and these curves only say where it stops succeeding, not what it was doing while it succeeded.

Negative and failed checks are listed here rather than in a footnote because the pre-registration named them in advance. The truncation failure is the one that would change a number if fixed, and it would move the deepest L=12 points UP.

Against the previous best, on the one axis the two ensembles share

the unconstrained ensemble stops at rho=0.471; the fall ensemble reaches 0.333



The fall fix changes the GRAPH ensemble, so no fall cell is paired with any earlier checkpoint and the two cannot be McNemar'd; required reasoning depth is the axis they share. The old cells also got 750k gradient steps against the scan's 375k, so this is not a like-for-like win and none is claimed. Where they overlap the new cells track the old ones at equal L on half the optimization, and then carry on for another 1.77x in depth: the previous best, L=12 unconstrained, solved 0.948 at its floor (T=7.8) and was never measured past it, while L=12/240k fall solves 0.985 at T=7.2 and 0.755 at T=13.8.